3.2 AUDIO SYSTEM

Some radio systems available on the 2005 TJ communicate on the PCI Bus. They use the bus for three reasons. The first is to communicate trouble codes, second is to receive dimming information, and the third is to receive cabin equalization information. The audio system is available in a 4 speaker base system and a 7 speaker system with an external subwoofer.

When troubleshooting output shorts or "output" error messages, the following applies:

On radios without an external amplifier, the term output refers to the path between the radio and the speaker. This type of circuit can be monitored all the way through the speaker connections by the radio assembly. When the radio displays a shorted output DTC with this type of system, the speaker, radio, or wiring could be at fault.

On radios with an external amplifier, the term "output" refers to the circuit between the radio connector and the amplifier. The radio is capable of monitoring only this portion and can tell nothing about the circuit between the amplifier and the speakers. Consequently, a shorted output DTC on this type of system would only refer to this circuit. A faulty speaker could not cause this DTC.

3.3 ELECTROCHROMIC COMPASS/TEMPERATURE MIRROR WITH LIGHTS

DESCRIPTION

The optional self-dimming Electrochromic Compass/Temperature Mirror has a vacuum fluorescent (VF) display that is integrated into the rear view mirror. The Compass/Temp Mirror includes the compass/temperature display and two map/reading lamps. This display provides the outside temperature and one of eight compass headings to indicate the direction the vehicle is facing. The Compass/Temp Mirror displays the compass heading and the outside temperature at the same time. The Ambient Temperature Sensor monitors the outside temperature and is hardwired to the Compass/Temp Mirror.

BUTTON OPERATION

The Compass/Temp Mirror incorporates 3 (mode/lamp) buttons to access and control various functions.

Left Button

- Press and Release
  - > Toggles Left Map Light On/Off
- Press w/ Right Switch for more than 5 seconds and Release
  - > Toggles Electrochromic Status On/Off -automatic On with each ignition cycle
  - > On mode is indicated by green status LED next to right button

Center Button

- Press for more than 3 seconds and Release
  - > Cycle through display status
    - – Compass / Temperature (Fahrenheit) - °F mode stored to memory
    - – Compass / Temperature (Celsius) - °C mode stored to memory
    - – Display Off - off mode stored to memory
- Press and Hold for 3-6 seconds
  - > Activate zone variance mode (adjust compass for true north)
    - – ″Z″ and the currently programmed zone (1-15) will flash in display window
    - – Each press of the center button will increment to the next zone
    - – When proper zone is selected, wait 5 seconds and mirror display returns to comp/temp (see variance zone map for proper zone number setting)
- Press and Hold for more than 6 seconds
  - > Activate compass calibration
    - – ″CAL″ shows in display window until compass is calibrated (verify correct variance zone prior to initial or re-calibration)
    - – Drive vehicle in a slow circle for 1.5 revolutions to recalibrate compass

Right Button

- Press and Release
  - > Toggles Right Map Light On/Off

ELECTROCHROMIC (EC) OPERATION (auto dimming feature)

The automatic dimming feature detects forward and rear light conditions and adjusts the reflectance level of the mirror to eliminate unwanted glare by the use 2 photoelectric sensors. The feature can be disabled by depressing the Left and Right buttons together for more than 5 seconds. The disable command will only remain in effect until the ignition is turned off. The EC function will automatically return to ON with each ignition cycle.

When the forward sensor detects daytime conditions, the rear sensor is inactive and the mirror remains in a high reflectance state.

When nighttime conditions are sensed by the forward sensor, the rear facing sensor is active and detects glare from rearward approaching vehicles or other glare producing light sources. The mirror